🏫 Unit 3 Homework — Equations & Inequalities

Complete independently · One/Two-Step · Distributive · Multi-Step · Special Cases · Inequalities · Word Problems

A. One- & Two-Step Equations

Problem 1

Solve: $x - 6 = 11$

17

Problem 2

Solve: $5x = 45$

9

Problem 3

Solve: $2x - 7 = 9$

8

Problem 4

Solve: $\frac{x}{4} + 5 = 8$

12

Problem 5

Solve: $-3x + 2 = -10$

4

B. Distributive Property & Variables on Both Sides

Problem 1

Solve: $4(x - 2) = 20$

7

Problem 2

Solve: $2(3x + 1) = 26$

4

Problem 3

Solve: $6x + 5 = 3x + 20$

5

Problem 4

Solve: $8x - 1 = 5x + 17$

6

C. Multi-Step Equations

Problem 1

Solve: $2(x + 5) = 3x - 1$

11

Problem 2

Solve: $6x + 2 = 3(x + 8)$

$\frac{22}{3}$

Problem 3

Solve: $10 - x = 2x + 1$

3

D. Special Cases — No Solution or Infinite Solutions

Problem 1

Solve: $5x - 2 = 5x - 2$ — how many solutions?

infinitely many solutions

Problem 2

Solve: $3x + 7 = 3x - 4$ — how many solutions?

no solution

Problem 3

Solve: $2(x + 6) = 2x + 12$ — how many solutions?

infinitely many solutions

E. Linear Inequalities

Problem 1

Solve: $x - 5 < 2$

x < 7

Problem 2

Solve: $4x \geq 32$

$x \geq 8$

Problem 3

Solve: $-5x < 20$ (careful!)

$x > -4$

Problem 4

Solve: $2x + 7 \leq 19$

$x \leq 6$

Problem 5

Solve: $-4x - 3 > 9$ (careful!)

$x < -3$

F. Word Problems

Problem 1

A phone plan costs $20 plus $0.05 per text. If the bill was $35, how many texts were sent?

300

Problem 2

You have $60 and want to buy shirts that cost $12 each. What's the max number of shirts you can buy?

5